INCIDENT REPORT

A DNS Spyware Alert, Traced to the Exact Browser Process

Process attribution by joining firewall and EDR telemetry, and the 16.5-minute clock skew that almost blamed the wrong browser

Item	Detail
Case	Adware DNS beacon attribution (fictionalized portfolio edition)
Endpoint	NB-FIN-207, Finance department
User	sinta.w
Detection	NGFW DNS signature in TAP (monitor-only) mode, read in the SIEM
Report date	Day 0
Prepared by	Tri Wibowo Cahyo, Security Analyst
Verdict	True Positive, low severity. Adware component inside the managed Chrome browser
Classification	Portfolio edition. Identifiers, domains, ports, counts and dates are fictionalized

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> **About this edition.** This report is based on a real investigation I led as a SOC analyst. For portfolio use, the organization, hostnames, usernames, IP addresses, the adware domain, source ports, event counts and dates are fictionalized. The investigation method, the tool behavior and the reasoning are kept as they happened. Times are local time on the day of the alert (Day 0).

1. Executive Summary

A firewall DNS signature flagged a Finance laptop asking for a known adware domain. The question was which program on the laptop made the request. The obvious suspect, an unmanaged Opera browser installed by the user, turned out to be innocent. **The requests came from the company-managed Chrome browser**, most likely through an extension in the user's profile.

Getting there needed two things. First, joining two logs that each hold half the answer: the firewall knows the domain and the source port, the EDR knows the process and its local port. Second, noticing that the SIEM's timestamp for these firewall events ran **about 16.5 minutes behind** the firewall's own clock. On the SIEM time the join failed and pointed at the wrong browser. On the device time, all three source ports matched `chrome.exe` in the same second.

Verdict: True Positive, low severity. Adware, not an attacker-controlled implant: it only runs while the user's browser runs and retries with exponential backoff when it fails.

Key findings

Aspect	Finding
What	Repeated DNS requests to an adware / browser hijacker domain from NB-FIN-207
Pattern	More than a thousand requests in 30 days, working hours only, two bursts per day, exponential backoff from 15 seconds up to about 10 minutes
Who asked	`chrome.exe`, the managed Chrome install under Program Files. Confirmed by a 3-of-3 source port match
Red herring	A user-installed Opera browser in AppData. Present and noisy, but absent at the true moment of the request
Clock finding	SIEM ingestion time for this firewall log source ran about 16.5 minutes behind device time, consistently
Telemetry limit	The EDR samples UDP DNS connections, so a process missing from EDR data is not proof it was idle

Top recommendations

1. Clean the Chrome profile on NB-FIN-207: review and remove unapproved extensions, then reset the profile if the requests continue.
2. Enforce a Chrome extension allowlist through browser policy, so the same adware cannot be installed again.
3. Remove the unmanaged Opera install as separate software hygiene, not as the fix for this alert.
4. Correlate firewall events with endpoint data on device time, not SIEM ingestion time, until the clocks are aligned.

2. Scope and Evidence

The investigation covered one laptop in depth and the whole fleet for scoping. Two independent data sources were used, and neither could answer the question alone.

Source	What it gave	Limit
NGFW DNS signature, read in the SIEM	Which laptop asked, which domain, source port, device time	Sees traffic from outside the laptop, so it cannot name the process. Runs in TAP mode, so it records and does not block
Cortex XDR network and process events	Which process opened each connection, its local port, its image path	Records connections to port 53, not the domain names asked. UDP DNS connections are sampled
SIEM, 30-day history	Volume, rhythm, other affected hosts	Its own timestamp for this log source is skewed (Section 8)

3. Timeline

All times are on Day 0, a Monday. The investigation took most of the morning.

Time	Event
09:00	I close an earlier EDR case on NB-FIN-207 as a false positive. The data I pulled shows only browsers talking to Google, WhatsApp Web and CDNs
09:12:40 (SIEM time)	New alert: DNS spyware signature for `pixelharborscan[.]info` from the same laptop. Device time on the event: 09:29:05
Morning	I reopen the closed case. Its data held no DNS information, so the closure did not answer this question
Morning	Scoping over 30 days: a long-running, session-bound pattern on this laptop, plus a few other laptops that went quiet weeks ago
Morning	Attempt 1, timing: Opera looks guilty
Morning	Attempt 2, source port join on SIEM time: zero matches
Late morning	Clock test: device time puts 190 of 190 hits inside browser activity; SIEM time leaves 86 outside it
Late morning	Source port join on device time: 3 of 3 ports match `chrome.exe`. Case attributed

4. The Alert and the First Closure

The alert came from a next-generation firewall that watches traffic in TAP mode: it sees a copy of the traffic and logs what it recognizes, but it cannot stop anything. It recognized the laptop asking the internal DNS server for `pixelharborscan[.]info`, a domain with a reputation as a browser hijacker (adware that changes the homepage, injects ads and rides along with the browser).

A few minutes earlier I had closed a different case on the same laptop as a false positive. That closure was correct for the data I pulled, which was process and connection events for the browsers. But that data held **no DNS names at all**. The closure answered "are the browsers talking to anything bad by IP?", not "did anything on this laptop ask for a bad domain?". I reopened it.

> **Lesson:** a closure is only valid for the questions its data can answer. When the data has no visibility into the thing a new alert is about, the earlier closure says nothing about that alert.

5. Scoping: a Persistent, Session-Bound Pattern

Before chasing the process, I checked whether this was one odd request or a pattern. Thirty days of SIEM history answered that it was a pattern.

- NB-FIN-207 asked for the domain **more than a thousand times** over the month, and was still doing it.
- A handful of other laptops had also asked for it, but all of them had been silent for weeks.
- The internal DNS servers appeared in the results too, but only because they forward queries upstream. They are relays, not sources.

The rhythm told me what kind of thing I was dealing with. Requests came in bursts: a failure, a retry after 15 seconds, another after 15 seconds, then 1, 2, 4 and 8 minutes, capping at about 10 minutes. That is **exponential backoff**, the way software retries a server that keeps failing. There were two bursts per working day, one in the morning and one around midday, and **nothing on weekends**.

Signal	Adware riding the browser	Attacker-controlled beacon
When it runs	Only while the user's session and browser run	Usually around the clock, including idle nights
Retry shape	Backoff after failures, bursts tied to browser start	Steady interval with jitter
This case	Matches	Does not match

That set the severity: a stubborn nuisance, not a breach. What remained was finding where it lived.

6. Attribution, Attempt 1: Timing

On the EDR, the laptop had the company Chrome install and, next to it, an **Opera browser installed by the user** in `AppData`. Not deployed by IT, not in the asset list, and the noisiest process on the machine.

I lined up the minutes around the alert: which processes were talking to the internal DNS server? Opera was there. So was Chrome, but Opera was far busier. An unmanaged browser meeting an adware domain made a tidy story.

Too tidy. When I compared every hit in the SIEM against browser activity on the EDR, two things did not fit. On one day the first hit landed about **six minutes before any browser was running**. On other days, hits fell in minutes when both browsers were silent. A browser cannot ask for a domain before it starts. The timing story was starting to crack, and a new theory appeared: maybe a background process, not a browser, was the trigger.

7. Attribution, Attempt 2: Port Join on SIEM Time

Timing alone could not separate two busy browsers. I needed something closer to a fingerprint.

Every outgoing connection gets a temporary **source port** from the operating system. It is close to random, so a source port plus a timestamp is nearly unique. Both logs record it: the firewall event carries the source port, and the EDR records the local port of each process connection. If the numbers match in the same second, the process is identified.

The alert at 09:12:40 (SIEM time) carried three source ports: 51877, 49302 and 57114. In the EDR window around 09:12:40 I found only Opera connections to the DNS server, and **none of their ports matched**.

Two lessons came out of this failure. First, the EDR does not record every UDP DNS connection; it samples them. So "process X does not appear" never proves "process X was idle", and I stopped using absence as evidence. Second, zero overlap across three ports was suspicious in itself. Sampling explains a missing match, not a total miss.

8. Which Clock Is Right?

Each firewall event in the SIEM carries two times: the **SIEM time**, when the SIEM received and processed the log, and the **device time**, the firewall's own clock when it wrote the log. On the alert they differed: 09:12:40 versus 09:29:05.

I first took that for ordinary delivery delay. But across every event for this domain the gap was **always about 16.5 minutes**, give or take a few seconds. Delivery delay is never that steady. One of the two clocks was simply off.

To decide which, I needed an independent reference, the way you check two wall clocks against the school bell. My bell was the EDR: browser activity timestamped by the laptop itself. I mapped all 190 hits from four working days against the windows when a browser was active, once with each clock.

Clock used	Hits inside browser activity	Hits outside browser activity
SIEM time	104 of 190	86 of 190
Device time	**190 of 190**	**0**

The device time was right, and the SIEM time ran about 16.5 minutes late. Every earlier oddity fell into place. The hit "six minutes before any browser" was an artifact of the late clock; on device time, the first hit each day came seconds to a few minutes **after** the browser opened, which is exactly how an extension that starts with the browser behaves. The "background process" theory collapsed. The suspect had not been strange; the witness's watch had.

9. Attribution Resolved

With the clock corrected, the port join from Section 7 had been run against the wrong minute: 16.5 minutes too early. I ran it again at the device time, 09:29:05.

All three source ports, 51877, 49302 and 57114, appeared in the EDR in the **same second**, on the same process: `chrome.exe`, running from the standard managed path under `Program Files`. Opera did not appear in that window at all.

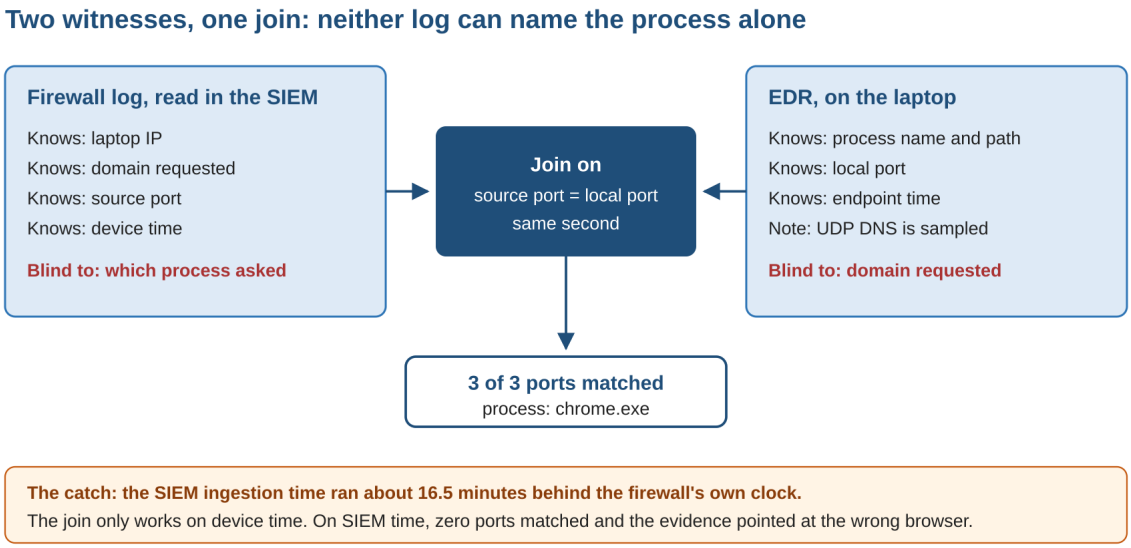


Figure 9.1 How the attribution works: two partial logs joined on source port and device time

Example queries, generalized. Checking the skew per event in the SIEM (AQL):

```
SELECT DATEFORMAT(starttime, 'yyyy-MM-dd HH:mm:ss') AS 'siem_time',
       DATEFORMAT(devicetime, 'yyyy-MM-dd HH:mm:ss') AS 'device_time',
       (devicetime - starttime) / 1000 AS 'skew_seconds',
       sourceport AS 'src_port'
FROM events
WHERE UTF8(payload) ILIKE '%<domain>%' AND sourceip = '<laptop-ip>'
ORDER BY devicetime ASC
LAST 7 DAYS
```

The port join in Cortex XDR (XQL), with the time picker set to the device-time window:

```
dataset = xdr_data
| filter agent_hostname contains "<hostname>"
| filter action_remote_ip = "<dns-server-ip>" and action_remote_port = 53
| filter action_local_port in (<port1>, <port2>, <port3>)
| alter process = actor_process_image_name, process_path = actor_process_image_path,
        local_port = action_local_port
| fields _time, process, process_path, local_port
| sort asc _time
```

10. Verdict and Recommendations

True Positive, low severity. An adware component runs inside the managed Chrome browser on NB-FIN-207, most likely as an extension in the user's profile. It starts with the browser, tries to reach its server, fails, and retries. No sign of an attacker-controlled implant, data theft or spread.

Had I stopped at the first suspect, the case would have closed neatly and wrongly: Opera uninstalled, the adware still in Chrome, and the same alert back the following week.

10.1 This laptop

1. Review the Chrome extensions in the user's profile and remove anything unapproved. Reset the profile if the requests continue.
2. Confirm the fix by watching for the same DNS signature from the laptop over the next working days.
3. Remove the unmanaged Opera install as software hygiene. It is a policy issue, not the cause of this alert.

10.2 Fleet and process

- Enforce a Chrome extension allowlist through browser policy.
- Align the SIEM's time handling for this firewall log source. Until then, correlate firewall events with endpoint data on device time.
- Add the source-port join to the team's playbook for DNS-based alerts, together with the clock check that must come first.

11. Indicators and ATT&CK Mapping

Indicator	Notes
`pixelharborscan[.]info` (fictionalized)	Adware / browser hijacker domain
DNS requests in exponential backoff bursts, working hours only	Behavior of a failing component tied to the browser session
First request each day shortly after browser start	Points to a browser extension rather than a service

Tactic	Technique	Confidence
Persistence	T1176 Browser Extensions	Strong inference: attributed to the browser process and starts with it. The specific extension was not identified in this report

12. Limitations and Lessons Learned

12.1 Limitations

- **No DNS names in the EDR.** Process attribution depended entirely on the port join.
- **Sampled UDP telemetry.** Missing EDR records could not be used as evidence of inactivity.
- **Extension not named.** The report attributes the requests to the Chrome process; naming the extension needs a look at the profile on the endpoint.

12.2 Lessons Learned

1. **Check the clocks before correlating systems.** Never trust one timestamp because it is convenient. Test both against an independent source.
2. **Absence of evidence is not evidence of absence.** Know what your tool records before reading meaning into an empty result.
3. **Prove the trace, not the look.** The most suspicious-looking program was not the culprit. The evidence decided, not the appearance.
4. **Rhythm tells you severity.** Backoff bursts that follow the user's working hours separate adware from a beacon before you even find the process.
5. **A closure is scoped to its data.** Reopen a case when a new alert asks a question the old data could not answer.